

Evaluating the Impact of Sanctions: Russia's Economy and Global Trade Dynamics

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Introduction

Sanctions are restrictive economic measures that countries take to deter and condemn the unjust violence of other states. Being widely used during the Cold War by the two blocs, after its end in 1989, the unification of the West and the East gradually lifted the sanctions and enhanced economic integration and interdependence. In theory, successful sanctioning assumes that the negative impact on the sanctioned state enforces constructive negotiations with the reversal of actions. This approach was effective against governments that depended economically on foreign investment, and for countries that did not rely on exporting natural resources. After the annexation of Crimea in 2014, and Russia's invasion of Ukraine, Russia faced more than 18,000 sanctions and became the most sanctioned country in the world. These events marked a turning point in policy design, as the sanctioned country was overly connected to the world. Russia ranks among the top exporters of crude oil, natural gas, coal, and other natural resources; its isolation would have a detrimental impact on global energy prices. This fact highlights the necessity for a thorough cost-benefit analysis to be made by policymakers to inflict the maximum damage on the Russian economy while minimizing their impact on Western countries and their allies.

This research aims to examine the effectiveness of the imposed sanctions, focusing on their impact on the Russian macroeconomic outlook, and the evolution of sanction policies throughout the conflict. The study highlights the universal measures that global governments take against regimes violating territorial integrity and examines the potential of economic measures to prevent military escalations.

History of the sanctions on Russia (2014-2024)

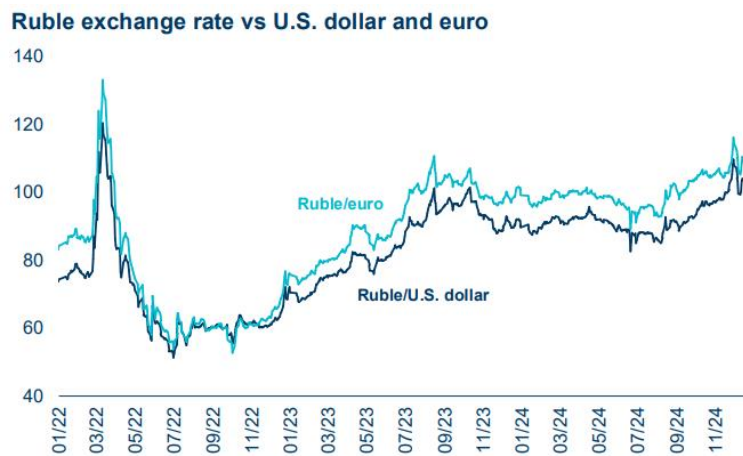
The first sanctions were imposed on Russia after the annexation of Crimea in 2014 by the US and the EU, displaying political disagreement with the actions of the Russian government, and condemnation of them. The sanctions contained diplomatic measures and asset freezes that were considerably indicative rather than being damaging for the Russian economy or isolating it from the global financial system. The domestic response to these measures was an increased demand for foreign currency that caused a 94% depreciation of the Russian ruble in the first post-crisis months. As opposed to 2022 the Bank of Russia did not impose substantial regulations on currency withdrawals and exchange rates in 2014, which directly amplified the impact on markets. Nevertheless, as the Russian economy did not face other risks, the records of their impact made a false impression of the softness of the Western countries, and of the lack of determination in these measures. The next wave occurred in 2017, when Russian proxy rebels in Eastern Ukraine shot down a Malaysia Airlines passenger plane, alongside multiple cybersecurity violations conducted by Russian hackers. This chain of sanctions negatively affected foreign direct investment in Russia and raised awareness among markets and investors about foreign sanction policies (McLean, Ryu, & Whang, 2023), but still could not backtrack military aggression. However, experts agree with the notion that the restrictions before 2022 were a measure to stimulate behavioural changes rather than economic destruction (Itskhoki & Ribakova, 2024).

Until February 2022, before the invasion of Ukraine, Russia was the third most-sanctioned country after Iran and Syria because of its actions in Ukraine in 2014. In 2022, it became the most sanctioned country, with more than 18,000 sanctions imposed. The sanctions were mandated mainly by the US and the EU; they were supported by many emerging powers at different times throughout the conflict. Kutlay and Öniş (2024) define these states as ‘actively neutral’ and

‘strategically autonomous,’ implying the pragmatic actions of the Global South countries in this conflict. Including active BRICS members, most of these countries diplomatically supported the sanctions without further enforcement. The shifting balance between democracies and autocracies has revealed inconsistencies between the political and economic stances of emerging powers such as China, India, Saudi Arabia, Brazil, and Turkey (Kutlay & Öniş, 2024). While condemning the military actions of the Russian government and supporting the territorial integrity of Ukraine, these states play a key role in Russia’s sanction evasions, and the ‘parallel’ import of sanctioned goods by foreign companies into Russia, diminishing the efficiency of sanctions.

Following the chronology of the sanctions and restrictions on Russia is crucial for evaluating their efficiency and dynamics over time. According to Itskhoki and Ribakova (2024), there are three main types of restrictions: financial sanctions, export-controlling sanctions, and trade embargoes. These measures aim to take a strategic approach to sanction policy by minimizing the losses to imposing countries and maximizing the damage to the Russian economy. The scope of sanctions has shifted primarily to natural resource exports to diminish fiscal revenues. The restrictions on the financial system slow down the overall trade of the sanctioned country by imposing limitations on international borrowing, and transactions involving foreign assets. They serve as a tool to cut off foreign currency revenue gains, increasing the transaction costs to the exporter. In a complete financial sanction framework, the country is excluded from any inflows of capital and revenues from exports. Therefore, the economy becomes reliant on barter exchanges and direct export-to-import swaps, excluding the country from the international financial system. Similar measures were applied to the Soviet Union during the Cold War, significantly constraining its growth potential. After February 2024, the US, EU, and G7 countries imposed financial sanctions such as freezing \$300 billion of Russian assets in Western countries, restricting access

to foreign currency markets, broader international borrowing restrictions, excluding most Russian banks from SWIFT, and limiting the functions of financial authorities (Bank of Russia, Ministry of Finance of Russia). Some Russian banks have yet to be removed from SWIFT, insulating the Russian economy from financial shocks due to a ‘loophole’ in foreign currency export revenue collection. This has further implications for currency dynamics and trade revenue efficiency that facilitates achievement of the Bank of Russia’s goals such as reducing inflation, financial stability, and economic growth. The recent removal of Gazprombank from SWIFT in November 2024 is estimated to limit budget revenue inflows significantly and has already initiated the trend of ruble depreciation over a few months by 20% (Figure 1, August 2024-December 2024, Hilgenstock, Pavytska, & Ivanchuk, 2024, p. 27).



Source: Bank of Russia, KSE Institute

Figure 1, Hilgenstock, Pavytska, & Ivanchuk, 2024, p. 27

By imposing export restrictions (for the firms supplying goods to Russia), Western countries focused mainly on prohibiting dual-use technologies that could benefit both military expansion and the supply of intermediary goods used in the production of the military-industrial complex. The scope of the sanctions widened over time and included constraints on the energy

sector, high-tech innovations, and other industries that improved Russian economic, military, and technological capacity (Itskhoki & Ribakova, 2024).

The third measure that determines Russia's fiscal revenues is sanctions on Russia's energy exports. Itskhoki and Ribakova (2024) suggest that the most damaging sanctions were imposed in 2023 by the EU, as it was the primary importer (Figure 2, January 2022, Hilgenstock, Pavytska, & Ivanchuk, 2024, p. 5), and was significantly reliant on Russian energy supplies. On February 5, 2023, sanctions included a complete ban on all oil products imported from Russia, and the further termination of natural gas, coal, and metal imports. These measures restricted the EU shipping and insurance companies from participating in the logistics of Russian crude oil. The risk of an increase in energy prices and losses for the EU because of the decreased supply of Russian energy was mitigated by the active role of the US in withstanding potential oil price shocks and the EU implementing a gradual energy exporter substitution. These sanctions aimed not to exclude Russian energy from global markets but to impose high transfer and operational costs on the exporter side. The price cap mechanism was implemented at the beginning of 2023, G7 and the EU countries introduced a price ceiling for Russian natural resources so that Western companies could legally engage with Russian energy exports while generating lower profits for Russia. However, after the price cap was established, at least 64 uninsured and unregulated Russian 'shadow' tanker violations were documented, raising ecological concerns and the actual enforcement limitations.

The barrier to rapid implementation of measures on Russian exports for the sanction coalition was the distribution of costs associated with putting restrictions and the collective action problem. The data on Russian oil exports and budget revenues display a noticeable decrease caused by the price cap as well as the EU trading restrictions in Q1 2023 (Figure 3, Hilgenstock, Pavytska,

& Ivanchuk, 2024, p. 6), with adjustments made to these measures operationally in Q3 2023. The rapid adaptation to the sanctions reveals a limitation of sanction design. Some states could consciously weaken enforcement, as Kutlay and Öniş (2024) suggest, significantly decreasing the efficiency of sanctions. Naturally, self-interest motivates countries to seek benefits, and cooperate publicly or confidentially with the Russian authorities, distorting the initial statistics.

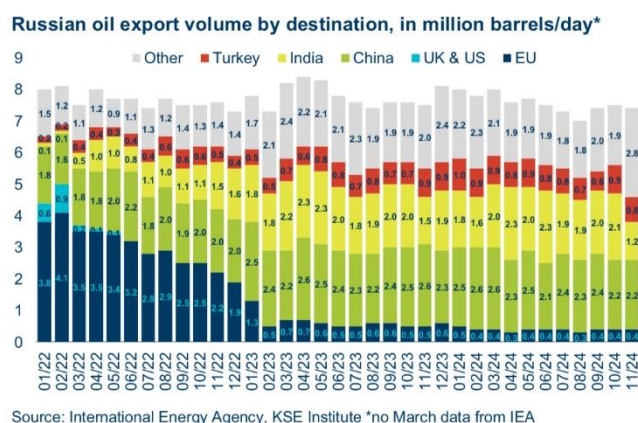


Figure 2. Hilgenstock, Pavytska, & Ivanchuk, 2024, p. 5

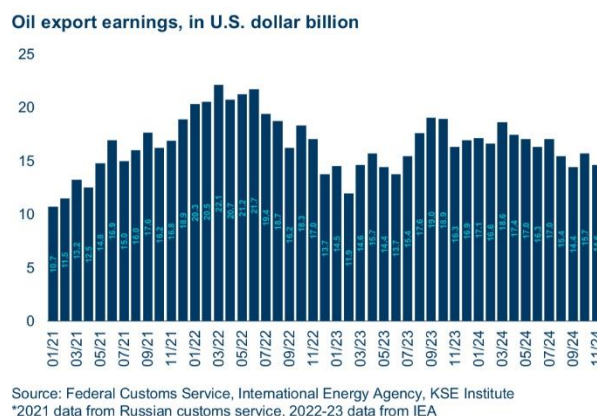


Figure 3. Hilgenstock, Pavytska, & Ivanchuk, 2024, p. 6

Impact of Sanctions on the Macroeconomic Capacity of Russia

The goal of the sanction coalition is the limitation of war financing, implying that the funding sources have to decline, and the key indicators of the Russian economy have to display a slowdown. The financial sanctions target the domestic exchange rate, aiming to create financial distress, and damage the current account balance. The sanctions on Russian imports set a goal to cause supply shortages, and limit consumption choices to create inflationary pressures on the economy. The gradual restrictions on Russian energy exports determine the budget revenues and, consequently, the war financing capabilities.

GDP, the primary economic indicator, may not have a meaningful influence on the macroeconomic evaluation of Russia. The growth decomposition relies on the military-industrial complex production, which cannot be a representative household welfare measure. The data display the stagnation in non-military retail sales and consumption as a part of aggregate output, still indicating economic growth (Figure 4, Hilgenstock, Pavytska, & Ivanchuk, 2024, p. 31). The reported increase in real wages is another example of fiscal stimulus driven by military recruitment, which is far from free-market pricing (Figure 5, 2022-2024, Bank of Russia, 2024). The unemployment level is at a record low because of the surging brain drain as the war started and due to partial mobilization in September 2022. The instability and threats to human rights violations have been key factors in the departure of qualified labour force cohorts. The increasing role of the government in investment and spending through interventions and fiscal injections over the last 3 years make the macroeconomic data incomplete. The government sector runs its own account on subsidized credit to strategically important industries based on the demand for military personnel and equipment on the battlefield.

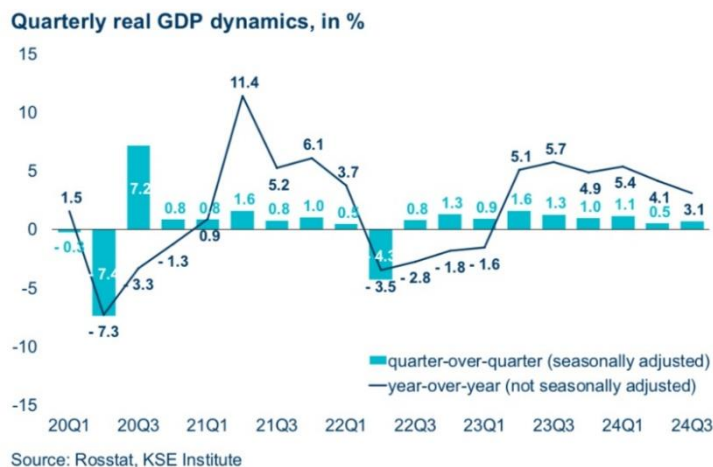


Figure 4. Hilgenstock, Pavytska, & Ivanchuk, 2024, p. 31



Figure 5. Hilgenstock, Pavytska, & Ivanchuk, 2024, p. 29

The implications of sudden financial restrictions imposed in the first weeks of the war drove the ruble exchange rate to its record of 120 rubles per US dollar in February 2022 (Figure 1, February 2022), indicating a panic that boosted the demand for foreign currency. This uncertainty caused liquidity deficits in the largest Russian banks and put pressure on the Bank of Russia's banking sector support mechanisms (Figure 6, Hilgenstock, Pavytska, & Ivanchuk, 2024, p. 25). The Bank of Russia quickly implemented a series of measures and market interventions after the war began. These included raising the policy rate from 9.5% to 20% during the February meeting of the Board of Governors, providing liquidity support to Russian banks, and temporarily shutting down the stock market. The other revenue-generating channel was the current account surplus in 2022; the data display a positive cumulative trade balance and current account in the first year of the war (Figure 7, 2022, Hilgenstock, Pavytska, & Ivanchuk, 2024, p. 13). The highest surplus of \$238 billion smoothed the negative impact of strict financial conditions, facilitating the recovery cycle of the Russian economy (Itskhoki & Ribakova, 2024). The adverse effect of tightening the import abilities (spending channel) and record amounts of energy exports provided excess foreign currency liquidity that appreciated the ruble against foreign currencies in the

following months (Figure 1, March 2022-August 2022). As the Russian budget revenues are highly dependent on energy exports, the increased supply of foreign currency through exports drove the ruble up. When discussing the ruble's market or fixed rate nature, it is necessary to note the effect of the currency decomposition of earnings on the exchange rate. The previous research of Itskhoki & Mukhin (2022) underlines the correlation between the two, evaluating the financial sanctions together with import restrictions as beneficial for ruble stability. After January 2023, the decrease in ‘non-friendly’ currency revenues (Figure 8, Hilgenstock, Pavytska, & Ivanchuk, 2024, p. 11) fell below other currency volumes, weakening the foreign currency liquidity of the budget, and reflecting similar trends in the depreciation of the ruble exchange rate over time (Figure 1, 2023-2024). The import restrictions decreased the foreign currency spending potential significantly, and eased the currency constraint further, triggering an abundance of foreign currency from exports in the Russian budget, and making military expenditure more feasible (Figure 9, 2022, Hilgenstock, Pavytska, & Ivanchuk, 2024, p. 10).

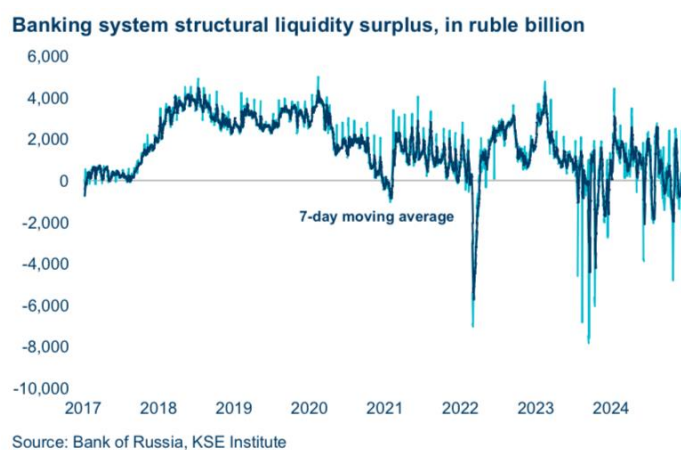


Figure 6. Hilgenstock, Pavytska, & Ivanchuk, 2024, p. 25

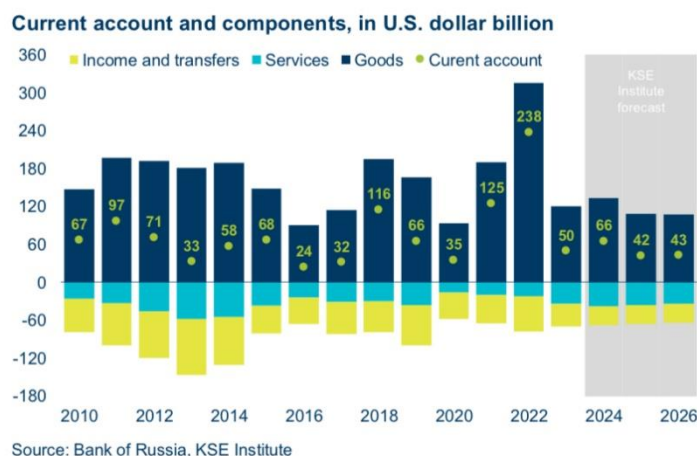
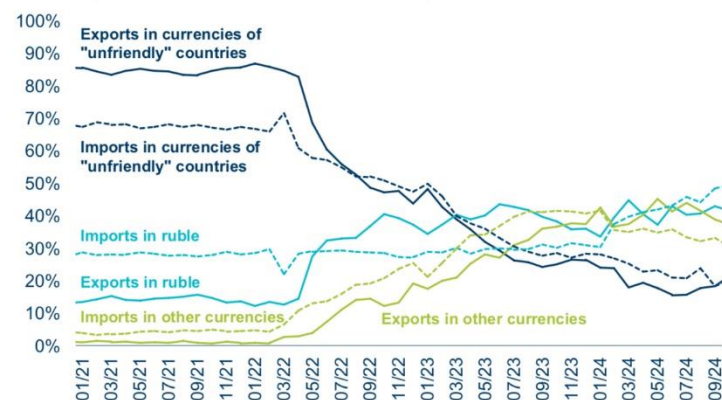


Figure 7. Hilgenstock, Pavytska, & Ivanchuk, 2024, p. 13

Currency composition of Russia's foreign trade, in %



Source: Bank of Russia

Figure 8. Hilgenstock, Pavytska, & Ivanchuk, 2024, p. 11

Monthly trade statistics, in U.S. dollar billion

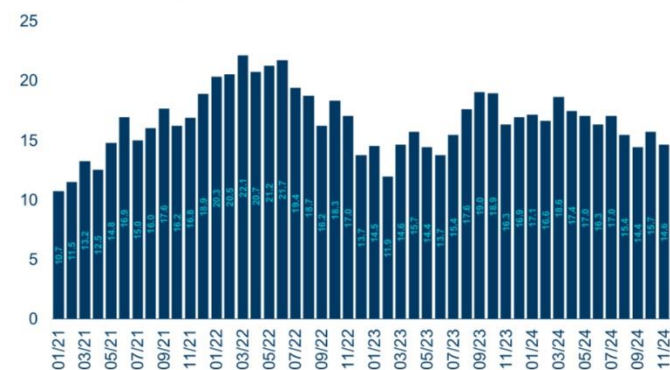


Source: Bank of Russia, KSE Institute

Figure 9. Hilgenstock, Pavytska, & Ivanchuk, 2024, p. 10

The price cap mechanism and the EU ban on Russian crude oil caused a 29% decline from \$58.2 billion in Q4 2022 to \$29 billion in Q1 2023 (Figure 10, Hilgenstock, Pavytska, & Ivanchuk, 2024, p. 6), decreasing by 50% in the current account surplus in Q1 2023 (Figure 11, Hilgenstock, Pavytska, & Ivanchuk, 2024, p. 9). This measure pressured Russian energy authorities to agree on lower profit exports to India to cover the volumes purchased by the EU in 2022.

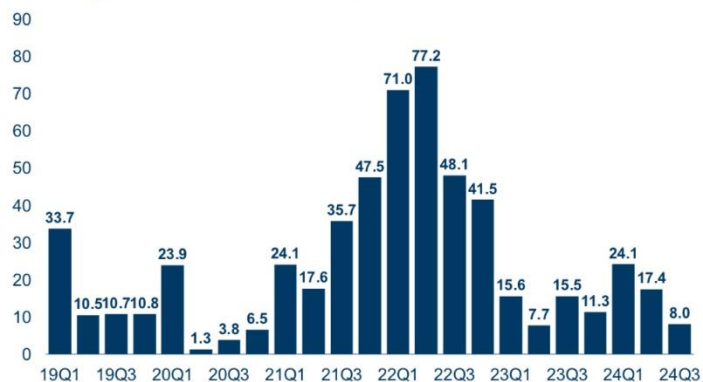
Oil export earnings, in U.S. dollar billion



Source: Federal Customs Service, International Energy Agency, KSE Institute
 *2021 data from Russian customs service, 2022-23 data from IEA

Figure 10. Hilgenstock, Pavytska, & Ivanchuk, 2024, p. 6

Quarterly current account balance, in U.S. dollar billion



Source: Bank of Russia, KSE Institute

Figure 11. Hilgenstock, Pavytska, & Ivanchuk, 2024, p. 9

In the long run, sanctions affected consumer prices significantly, causing a cumulative 30% increase during wartime (according to Governor Nabiullina's October Board of Directors meeting conference). Governor Nabiullina mentioned the increasing trend in inflation expectations, and the diminishing impact of the policy rate hikes on inflation in comparison to the previous rate hike cycle immediately after the beginning of the war in February 2022, when the Bank of Russia managed to handle the inflationary pressures by increasing the rates from 9.5% to 20%. The current austerity cycle with a policy rate of 21% has limited influence on decreasing inflation (estimated to be 9.5% by the end of 2024, according to Nabiullina), primarily due to an expansionary fiscal policy. The increasing budget deficits (Figure 12, Hilgenstock, Pavytska, & Ivanchuk, 2024, p. 15) and the record 40% allocation to military spending, with an additional \$15 billion (according to the Minister of Finance Siluanov) expansion to cover military expenditures in Q4 2024 indicates the policy divergence. The funds are channelled through subsidized credit with non-market low interest rates, which disrupt the credit market, and overheat the aggregate demand. These deficits are estimated to be compensated by the National Welfare Fund (NWF) – a state-owned investment fund formed by export surpluses, initially created to pay for the losses from the oil price volatility. Currently, it is used for war and fiscal deficit financing through withdrawals of liquid assets, as well as by OFZ (domestic debt) issuance (Figure 13, Hilgenstock, Pavytska, & Ivanchuk, 2024, p. 17). Budget expansion that is not sustainable due to decreased energy revenues compared to 2022 puts inflationary pressure and limits the impact of rate hikes by the Bank of Russia. The annual average fiscal stimulus of 10% of GDP burdens the Bank of Russia's cycle of policy rate hikes. In the October meeting, Governor Nabiullina indicated further policy rate increases, and the extension of the contractionary cycle (Figure 14, Hilgenstock, Pavytska, & Ivanchuk, 2024, p. 28). This divergence underscores the long-term impact of sanctions on the macroeconomic stability of

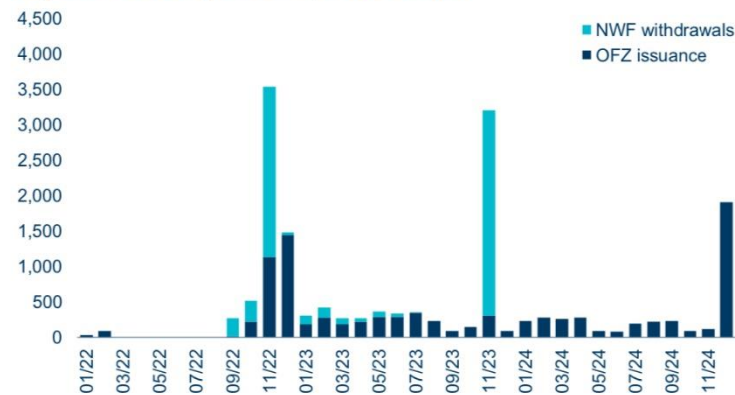
Russia, limiting budget and borrowing capacities while showcasing the contrast between militaristic fiscal expansion and monetary tightening in combating inflation.

Federal government balance, in ruble billion



Figure 12. Hilgenstock, Pavytska, & Ivanchuk, 2024, p. 15

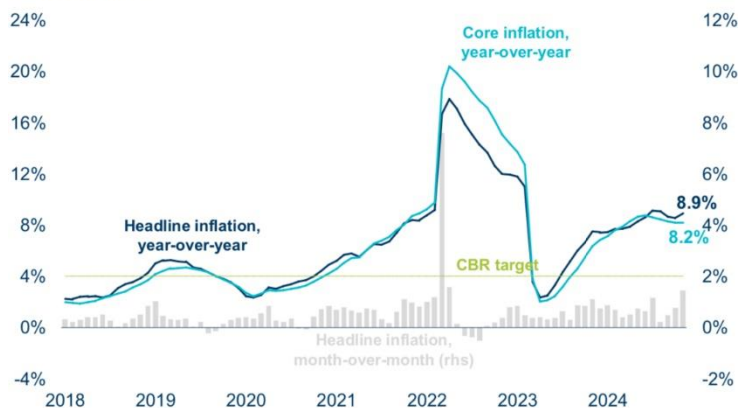
Key fiscal financing channels, in ruble billion



Source: Ministry of Finance, KSE Institute

Figure 13. Hilgenstock, Pavytska, & Ivanchuk, 2024, p. 17

Inflation, in %



Source: Bank of Russia, KSE Institute

Figure 14. Hilgenstock, Pavytska, & Ivanchuk, 2024, p. 28

Impact of Sanctions on the Global Trade and FDI

The theory of international trade suggests that for an open economy to be impacted by sanctions, it must be integrated into global trade and have interdependent ties for supply shocks to affect the target economy. However, decreasing interdependence diminishes the impact of sanctions, as the country becomes more isolated and reliant on domestic production due to severe restrictions. This conceptual paradox theoretically marks sanctions as an ineffective way of political stimulus and pressure. In real life, Russian energy resources cannot be excluded because of the scarcity of the resources but can be reallocated to emerging markets (China, India, Türkiye). As mentioned above, the sanctions on Russia are not universal but rather unilateral. The GDP contribution of the West to the world's economy declined from 75% to 60% in the post-Cold War period, resulting in a diminished isolation effect of unilateral sanctions that do not include the Global South countries. Given the high dependence of the Western countries on Russian energy and natural resources prior to the invasion (Figure 2, before 2022), the effects of sanctions without worldwide support have less impact than they had in the Cold War, making Russia a vital actor in world trade. Therefore, secondary sanctions enforcement and international cooperation determine the efficiency and applicability of the policies (Itskhoki & Ribakova, 2024).

In the years between the conflict stages in Ukraine, Russia had already faced a reduction in FDI in the three years before the annexation of Crimea in 2014, on average from \$58 billion annually to \$20 billion from 2014 to 2022 (McLean, Ryu, & Whang, 2023). Research on the short-run and long-run effects of sanctions on FDI has contrasting results, displaying the necessity to incorporate many country-specific factors. Therefore, the observations will rely on the research of McLean, Ryu, and Whang (2023), with monthly-level data on foreign investment transactions in Russia. Economic sanctions indicate the political risk level of the country, and ongoing property

rights violations together with failures in the integrity of the judicial system impact the willingness of the MNCs to invest. The evidence from the recent sanctioning cycles shows the high risks of military conflicts in countries facing sanctions (Lektzian & Sprecher, 2007). After February 2022, the Russian economy faced a wave of strategic voluntary exits or suspensions of foreign corporations, affecting the exposure to imported consumer goods in Russia. McLean, Ryu, and Whang (2023) indicate that data display greater cautionary capital outflows with the US or EU-imposed sanctions due to risks involving businesses in the US and EU countries. Foreign Russian government debt holders have rapidly withdrawn their investments after February 2022, which indicated a contraction in overall government borrowing capacities in the first months of the war (Figure 15, Hilgenstock, Pavytska, & Ivanchuk, 2024, p. 19). By 2023, the leftover supply of OFZs was reallocated to Russian corporations for federal budget funding. These findings highlight the sanctions' impact on foreign companies' willingness to invest in the sanctioned country. Although the foreign capital and MNCs rapidly left Russia after the invasion of Ukraine, Russian authorities strategically fulfilled the industries by offering Russian businesses below-market interest rate credits and subsidizing their substitution of major foreign enterprises (such as McDonald's, Starbucks, IKEA, Shell, Renault, H&M, Zara, and others). This reallocation significantly shaped the decisions of Russian investors to maintain their businesses in Russia. Restrictions imposed by Western countries on Russian business owners made it impossible for sanctioned individuals to maintain enterprises abroad. With the foreign sanctions, the 'support' of the Russian government in providing considerable advantages for businesses delayed and smoothened the supply shock for consumer goods and the social uncertainty against a shortage of daily products and services at the beginning of the war. The matching of new suppliers and logistics in the market significantly mitigated inflationary shocks caused by supply disruptions.



Figure 15. Hilgenstock, Pavytska, & Ivanchuk, 2024, p. 19

The impact of the sanctions and restrictions related to Russia's actions in Ukraine on global trade dynamics is illustrated in Figure 2, as the shift of energy exports from the EU to China and India becomes perceptible throughout the post-war period. This suggests that trade channels are flexible and can transform even in the short to medium term. As a result of this substitution, Russia's energy export volumes have not changed significantly in the last three years. Hence, the export revenues and budget balances declined because of explicit sanction policies, contributing to financial losses. Government stimulus of parallel imports partly returned the sanctioned goods to the shelves, creating inflationary pressure due to higher intermediary transportation costs. This made the intermediary countries better off, and created a profit opportunity, ignoring the ethical aspect of the action. This reduces the effectiveness of sanctions and leads to deviations from the original calculations by the policymakers.

Policy Recommendations

This research is mainly on various sanction types and their enforcement. The broader sanctions (ex., SWIFT, crude oil bans) have a diminishing impact on Russia's long-term macroeconomic stability after the market adjustment and impose high administrative costs on the sanction coalition. The lack of enforcement due to the immense connections between the international financial and trade systems creates opportunities for sanction evasion and shifts towards substitutes, mainly in the Global South countries. An example of that would be the transfer of crude oil export volumes from the EU to China and India, along with SWIFT payments for Russian energy made through unrestricted banks, and Chinese payment systems. These measures still put pressure on and impose additional costs on the sanctioned economy, but their substitution is possible in the short-to-medium term and cause adverse shocks to energy price stability by a short-term supply cut. On the contrary, export-specific sanctions, such as enhancing the price cap mechanism, are proven to be manageable in regulation and enforcement by targeting a specific sector and revenue channel.

Since the price cap enforcement in early 2023, the losses for Russian oil and gas revenues have been estimated to be \$114 billion, tightening the budget revenues by 24% in 2023 compared with 2022 statistics. The result of a 33% decrease in energy revenues for Russia was later compensated through price cap violations by using the 'shadow fleet,' highlighting the direct impact of sanction enforcement on the losses faced by the restricted state (International Working Group on Russian Sanctions, 2024). Therefore, the creation and cooperation of sanction policymaker agencies such as the US Treasury Department's Office of Foreign Asset Control (OFAC), Office of Financial Sanction Implementation (OFSI) in the UK, and country-specific EU structures responsible for tracking the efficiency of sanctions could lead to an improved evaluation

of policies, and consolidated enforcement and regulation mechanisms. The strong stance on imposing the price cap could achieve crude oil supply stability and lower the profit margins for Russia at the same time.

Shipping companies violate the price cap regime by providing services to Russian companies at above-price cap margins, thereby increasing Russia's energy revenues. Regulating this sector through firm-specific audits and obligatory reports could improve sanction enforcement, and lower deviations from the base model of the price cap. Austere measures that limit the further activity of sanction-violating firms in the sanctioning states could stimulate compliance with the policies.

The overall enhancement of administrative functions in sanction design within the governments of sanctioning countries, coupled with the widening of the sanction coalition, could strengthen the prudence of the policies. Coalition expansion can be achieved by providing favorable trade conditions to 'neutral states' that seek economic privileges by trading with Russia. Multilateral cooperation would improve sanction efficiency, as it limits 'loopholes', and prevents policy evasion.

The global energy market conditions and the political environment will impact the supply of discounted Russian crude oil to the market. In recent months, a decline in Chinese energy demand, along with collaboration among OPEC+ members in a gradual oil supply increase, could stimulate the future price cap decrease (International Working Group on Russian Sanctions, 2024). This will reduce Russia's budget revenues from energy exports. The mechanism is estimated to work unless the price cap is set above the below-market average price charged to India and other countries to guarantee oil supply reallocation in February 2023 after the EU crude oil import restrictions.

Conclusion

The debates on the short-term and long-term consequences of sanctions in the case of Russia were resolved through empirical evidence that in the short run, the supply shock hit the economy and raised commodity prices, while in the longer run, the supply chains adjusted through China, India, Türkiye, and post-Soviet countries by ‘parallel import’ (Itskhoki & Ribakova, 2024). As energy exports were transferred to China, India, and Türkiye, the relative prices started to stabilize, with the help of the ruble appreciation in Q2 2022. This caused the trade balance to shift with strict import restrictions and adjusted high foreign currency revenues from energy exports. The same applies to the EU economy, where the estimated recession at the end of 2022 did not occur despite the significant shift towards new energy suppliers by 2023. The higher-than-expected elasticity of substitutions led to a smoother switch in trade partners at acceptably low costs for both sanctioning countries and Russia. China and India mostly covered the 5 million barrels a day of oil left by the EU+G7 countries in the first months of the conflict, protecting the Russian economy from substantial export losses, with China importing Russian resources at nearly market prices. The graphs display declining energy revenues with quantities comparable to those in the pre-war period, proving the tangible negative impact of export sanctions on the Russian economy. Therefore, a decrease in the market oil price can harm revenues significantly; discounted high prices, as in the case of Russian exports, generate more profits than lower prices at a discount regardless of other variables.

Russia has been working on a ‘Fortress Russia’ approach since 2014 to withstand possible external financial pressures. Therefore, the data on the financial damage of sanctions within the last few years has revealed its weaknesses and limitations of self-reliance and economic mercantilism. Sanctions cannot serve as the ultimate measure of deterrence and adverse treatment

to stop conflicts, as they are, by nature, only tightening the initial budget constraint, not cancelling it out. As long as Russia's energy is in demand, the budget will continue to generate profits to finance the war. The conventional evaluation of policies in terms of success does not apply in the case of sanctions. Therefore, the assessment should include a cost-benefit analysis for the sanction-imposers and the subjects, together with a long-term strategy for systemic enforcement and policy design. The development of sanctions listed in the previous sections could serve as an extended model of already working sectoral sanctions. Reducing Russia's energy profits while keeping the energy supply at desirable levels could be the goal of future policymakers in eliminating global energy price shocks, with the gradual increase in the crude oil supply within OPEC+. Although, theoretically, the impact of sanctions can be predicted and evaluated, the lack of enforcement or realized logistic changeovers by the sanctioned country can determine the magnitude of the restrictions on the actual output (Itskhoki & Ribakova, 2024). These disruptions create a gap between the expected and realized damages of sanctions.

Limitations & Significance

Ongoing debates on Russian data transparency after February 24, 2022, began with the start of the war, when international audit companies withdrew from Russia, and Russia stopped its membership in international organizations such as the Council of Europe and the European Court of Human Rights. These debates reflect the opinion that Russian authorities use economic data as a weapon of war. The lack of institutional development of the Russian statistical agencies (primarily Rosstat), and their compliance with the Kremlin's narrative makes it possible for them to withhold information and manipulate data. In the second half of 2024, Rosstat stopped publishing data on crude oil and gas production statistics, detailed trade statistics, most of the

budget expenditure statistics, and the CPI basket components. Although the official data may have been falsified, this paper and the data derived from published research do not include any other forms of information (intelligence, insider information, and others) as they do not have enough credibility. KSE Institute's 'Russia Chartbook' contributed to the research with graphical data displaying Russian macroeconomic indicators.

The interdisciplinary approach of this research aims to combine the political and macroeconomic variables of both Russian and European decision-making processes to assess the sanction dynamics of the past three years. The snapshot of the current state of the Russian economy and its reactions to sanctions will shape the future of the military conflict. Considering the peace negotiation agenda set forth by President-elect Donald Trump, it is anticipated that the short-term financing capabilities of the war will significantly influence both the outcome of the negotiations, and the bargaining power of the involved parties in the conflict.

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